



Northern Bukidnon State College

eSAS: A Web-based Student Organization Information System with Club Membership Module

A Capstone Project

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CHAPTER 1

THE RATIONALE

Introduction

Club organizations are essential for creating a supportive community where students feel connected and engaged. These groups not only help students make friends but also build valuable skills like communication and leadership, which can enhance their academic performance and professional development (Muñoz et al., 2017). This sense of belonging is crucial, as it leads to more active participation and better overall academic results.

For these clubs to thrive, managing student records efficiently is key. Proper record-keeping ensures that information is accurate and up-to-date, which helps clubs operate smoothly and meet their goals. Yilmaz and Bozkurt (2018) highlight that well-organized data management is vital in educational settings, where handling large amounts of student information is a daily task. Without effective systems, clubs may struggle with tracking participation, evaluating performance, and meeting individual needs. Good record management also strengthens communication between students and administrators, helping everyone stay on the same page.

Research has shown that many student clubs face challenges with data management, such as lost records and delays in processing applications. Smith and Brown (2020) found that these issues often stem from outdated handling practices. Their study suggests that digital solutions, like web-based



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systems, can address these problems by centralizing data storage and management. Such platforms not only streamline administrative tasks but also boost overall efficiency by automating routine processes. This underscores the need for modern technology to improve how student organizations function.

In response to current issues, the researcher suggested the development of Student Affairs and Service called eSAS: Web-Based Student Organization Information System with Club Membership Module. This innovative system is designed to transform how student clubs manage records and engage with members. eSAS offers a central platform where students can easily register, view activities, and receive notifications. Moderators can use it to foster student involvement and manage club activities, while administrators can benefit from real-time data analytics and streamlined operations. With features like email notifications, a student profiling system, and a club ID generator, eSAS provides the tools needed for effective management. This system aims to enhance the student experience, improve communication, and ensure that student clubs are well-organized and prepared for the future.

1.2 Statement of the problem

The Northern Bukidnon State College - Student Affairs and Services (SAS) currently lacks a digital or electronic system for managing student organization records and information. This absence of a modern platform means that tracking, monitoring, and generating reports for student clubs is a major struggle. Without digital records, it's challenging to maintain an accurate



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history of club activities, manage memberships, and organize information effectively. The reliance on paper-based systems leads to inefficiencies and redundancy, making the process time-consuming and prone to errors.

As a result, administrators face significant difficulties in accessing real-time data and efficiently managing student records. The processes not only consume excessive time and effort but also hinder the overall efficiency of SAS operations. This lack of a centralized system impacts the ability to provide timely updates, address student needs promptly, and maintain organized records, ultimately affecting the smooth functioning and effectiveness of student organizations. Implementing a digital solution like eSAS could alleviate these issues by streamlining record management, enhancing communication, and improving administrative efficiency.

1.3 Objectives of the study

The aim of the study is to design and develop a web-based that enhances the management and communication of student clubs at Northern Bukidnon State College - Student Affairs and Services. Specifically, this study aims to achieve the following objectives:

1. To gather requirements in SAS administration through key informant interviews that are used for the analysis of data/ information for the design of the system.



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2. To design new process flow, UI/UX, and digital repository through diagrams, web frameworks, and DBMS for the overall design of the system.
3. To develop a sub-system under the existing SIS of the NBSC college through PHP, Javascript, HTML, CSS, JQuery, and MySQL for the overall development of the system.
4. To test the system using through User Acceptance Testing (UAT) that measures the system usability and ensures the software or system meets the user's requirements.
5. To implement the developed system through live hosting at Northern Bukidnon State College - Student Affairs and Services Office.
6. To evaluate the system through ISO 25010:2011 standards that ensure quality and effectiveness.

1.4 Scope and Limitation of the Study

The study focuses on the implementation of the eSAS system, a web-based Student Organization Information System with Club Membership Module for Northern Bukidnon State College (NBSC). This system is specifically designed for all students, moderators and SAS administrators at NBSC. The major components of the system include student profiling to maintain detailed records of each student's club memberships and participation history, email notifications to keep students informed about club events and updates, club ID generation for fast member verification and record-keeping,



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basic data analytics to visually represent and analyze data related to student organizations and club memberships, report generation to create comprehensive and accurate reports on various aspects of club activities, and a chat box for communication between moderators and students. Additionally, the system will integrate with the school's server to ensure smooth operation and provide an overview of the Student Body Organization and Central Student Government within each club.

While this study offers important advantages, it also comes with certain limitations. Firstly, the system be tested only with a stratified sample of students from selected departments at NBSC, specifically TEP, IT, and BSBA, which ensures representation across different departments but may limit the generalizability of the results to the entire student population. Secondly, the data analytics capabilities of the system are basic, providing only essential functions such as charts, graphs, and dashboards, without more advanced analytical features. Lastly, the effectiveness and efficiency of the system depend on the availability, such as a stable internet connection.

1.5 Significance of the Study

The significance of this study extends beyond the immediate benefits of developing a web-based system for managing student organization clubs at Northern Bukidnon State College (NBSC). In an era of rapid technological advancement and increasing demand for efficient educational management, the eSAS system represents a pivotal step towards modernizing and streamlining



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administrative processes within academic institutions. The following stand to benefit significantly from this project:

- **To Northern Bukidnon State College (NBSC).** The college will benefit from improved student engagement and a more vibrant campus life. The system can serve as a model for other departments and institutions, showcasing the potential of technology to enhance educational experiences.
- **To SAS Administration.** The administration will benefit from better management of student organizations and streamlined administrative procedures. The system will help keep accurate records and facilitate easy communication with students.
- **To NBSC Students.** The system will provide students with easier access to information about student clubs, enabling them to join clubs, stay informed about events, and actively participate in campus activities, thereby enriching their college experience.
- **To the Club Moderators.** Moderators of every club will find it easier to manage club activities, track member participation, and communicate efficiently with members. The system will assist in organizing events, maintaining member profiles, and generating reports.
- **To Future Researchers.** This study can serve as a reference for future research projects related to web-based management systems. Future researchers can build upon the insights, methodologies, and findings of



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this study to further innovate and improve student organization management systems.

Definition of Terms

1. **eSAS System:** A web-based platform designed to manage student organization memberships and related information. It includes various features such as student profiling, email notifications, report generation, and communication tools for the Student Affairs and Services (SAS) Office at Northern Bukidnon State College.
2. **Student Affairs and Services (SAS):** A department in Northern Bukidnon State College responsible for managing student activities, organizations, and services to enhance the overall student experience.
3. **Club Membership Module:** A component of the eSAS system that facilitates the management of student organization memberships, allowing students to register for clubs, receive updates, and track their participation.
4. **Student Profiling:** A feature of the eSAS system that maintains records of each student's club memberships, participation history, and relevant personal information to streamline administrative processes.
5. **Club Moderators:** Faculty or staff members responsible for overseeing the operations of student organizations, ensuring that clubs meet their objectives, and managing communication between students and the SAS Office.



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6. **ISO 25010:2011 Standards:** A set of international standards that define the quality requirements and testing criteria for software systems to ensure their functionality, reliability, usability, and effectiveness.

7. **Data Analytics:** Basic analytical functions integrated into the eSAS system, allowing for the generation of charts, graphs, and reports to visually represent student organization data and performance.

8. **ERD (Entity Relationship Diagram):** A diagram used in system design to represent the relationships between different data entities within the eSAS system, helping in understanding data structure.

9. **DFD (Data Flow Diagram):** A diagram that shows the flow of information within the eSAS system, detailing how data moves between different processes and entities.

10. **UML (Unified Modeling Language):** A standardized modeling language used to design the structure of the eSAS system, including system components, data interactions, and workflows.



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11. **PHP, JavaScript, HTML, CSS, JQuery, and MySQL:** Web development technologies used to design, develop, and maintain the eSAS system. These tools ensure the system's functionality, user interface design, and database management.

12. **Email Notifications:** A feature of the eSAS system that sends automatic messages to students and moderators about club events, updates, and important information regarding student organizations.

13. **Club ID Generator:** A feature within the eSAS system that generates identification for student club members to facilitate easy verification and record-keeping.

14. **Report Generation:** A function of the eSAS system that allows for the creation of detailed reports on club activities, member participation, and other relevant information for the SAS office and club moderators.

15. **Chat Box:** A communication feature in the eSAS system that facilitates real-time messaging between club moderators and students, enhancing coordination and collaboration.



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16. **Northern Bukidnon State College (NBSC):** A public institution of higher education in the Philippines, where the eSAS system will be implemented to manage student clubs and organization memberships.

17. **Stratified Sampling:** A sampling technique used in the study to ensure representation of students from different departments (TEP, IT, BSBA) within NBSC for testing the eSAS system.

18. **TEP, IT, BSBA:** Acronyms representing departments within NBSC – Teacher Education Program (TEP), Information Technology (IT), and Bachelor of Science in Business Administration (BSBA). These departments are part of the study's sample.



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Chapter II

REVIEW OF RELATED LITERATURE AND STUDIES

To expand the understanding of the study, related published materials were used, which are essential for expanding the knowledge base. These resources also serve as valuable guides for achieving the target objectives, as the proponents provide references to other related studies and allow for potential improvements.

As stated by (Cao, H., & Chang, M. et al., 2023), high school clubs serve an important role in allowing students to explore their interests and hobbies outside of the classroom. High school clubs occur in a variety of sizes, ranging from informal meetings of like-minded individuals to large-scale national organizations. However, the success of these groups in generating involvement and a feeling of community among members varies greatly. Despite its current functionality, the system has several limitations that can be improved upon. For instance, it lacks the ability for club members to communicate with each other and does not offer specific roles for users within their respective clubs. While these limitations do not significantly impede users' experiences, incorporating these features would undoubtedly enhance their overall satisfaction. This can create a suboptimal user experience and make the application appear unpolished. To address the current limitations of the system, future plans include incorporating features that enable communication between



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individual club members and offering specific roles for users within their respective clubs. These additions are expected to enhance users' overall satisfaction. By improving these aspects, the aim is to create a more refined and polished application that provides a seamless and enjoyable user experience for all club members.

(R. Vidya Pradosh et al., 2021) state that the Cricket Club Management System is entirely automated. The suggested system allows users to reserve field for certain days, seek club membership, and register for various training batches. This program makes it easier to maintain the club's database. Using this program will limit the margin of error and make it easier to regulate members. End users can register for membership, book ground, enroll in various training batches, and get management notifications. The admin performs managerial duties. Admin may authorize bookings made by end users, issue messages to the online notice board, and check various members of the club and registered users.

According to the study of (T. Liechty et al., 2020), members of leisure-based groups frequently have access to locations that foster a sense of community. As technology progresses, virtual communities are becoming more popular; however, research on virtual communities from a leisure standpoint is scarce. A study used netnography to investigate the meanings that members in a leisure-based virtual community assign to their experiences of community. The Hogwarts Running Club, a virtual club on Facebook, served as the study



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setting. Member postings were gathered on certain days throughout a six-month period. The dataset was freely coded, and the resulting coding scheme was compared to the four fundamental components of Sense of Community Theory. The findings corresponded well to the four-factor model. Participants cited a sense of belonging and connection to the community due to social support, shared experiences (both positive and unpleasant), discussions about the Harry Potter books and films, which shared symbols and vocabulary, and being “part of something bigger.”

Modernizing school services through technology significantly enhances the overall success of educational institutions by providing efficient and high-quality services to students, teachers, staff, and administrators (Balcita and Palaoag et al., 2020). These days, automated management systems have become necessary in schools and organizations that have to switch from using manual methods to using portable computing technologies (Srimathi R. et al., 2020). This change improves the management of student information and reports through the use of computer technologies and centralized databases. (Reddy and Rathna, 2020).

According to (Rohini et al., 2023), a college club management system can help organize and manage various components of student organizations, such as membership, events, communication, and records. In the study of (Cao and Chang et al., 2023), school organizations serve an important role in allowing students to develop their interests and hobbies beyond the classroom.



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Accordingly, extracurricular activities provide a space for certain interests that students can select based on their passions, abilities, and requirements. Under the guidance of an instructor or teacher, this type of activity typically allows students the freedom to express themselves through collaborations.(Kardiansyah, M.Y., & Qodriani, L.U. et al., 2018).

Colleges and universities are basically developed to provide educational services to the public, with the goal of equipping students with information, skills, and critical thinking abilities that will prepare them for a variety of jobs and life opportunities. These institutions have grown throughout time into complicated organizations that operate similarly to enterprises or industries. They cover a wide variety of processes and procedures that enable the smooth operation and effective delivery of services (Grepon, B. G.; Baran, N.; Gumonan, K. M. V.; Martinez, A. L. M. et al., 2021).

Using computers to manage student information provides multiple benefits, including fast searches, easy access to information, high reliability, huge memory capacity, confidentiality, and long-term data keeping (Vidanagamage et al., 2020). Stated in the study of (Jamis, Aziza, Jonoa et al., 2020), that Student Activities Management System aims to give a more reliable procedure, this is not compatible with the present paper-based technique, which currently in use. This study highlights the significance of offering a management information system for the institution so that both students and administrators can benefit from more informed and timely decisions and plans.



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In the modern educational landscape, fostering effective communication and collaboration among student clubs is essential for building a dynamic and engaged campus community (Chenjeri, A., T. Satya, K. Santhan Mourya, Muddada, M. K. et al., 2024). Student club development and management are vital components of educational institutions' missions. Clubs help students grow holistically by offering opportunities for them to pursue their interests, learn essential skills, and build a sense of community. This not only aligns with the institution's mission and purpose, but it also prepares students to be comprehensive, competent, and confident citizens capable of facing the world's challenges (Balcita and Palaoag et al., 2020). College students' supportive relationships with mentors—professors, advisers, and other caring adults to whom students turn as they explore their interests and career paths—are crucial to their growth and academic achievement (Raposa, E. et al., 2020).

In the college statistics accessible now multiple duties that are done manually in the management system. These are all paper-based activities that are costly and need a lot of time Different departments address various duties. This causes serious connecting issues data while preventing duplication (G. Murugan, V. Hemalatha, P. Gomathi et al., 2020). With using the management information system that can assist educators and staff in organizing, implementing policies, and assessing each activity in school. Given that information technologies were mostly used in educational institutions, helping



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to oversee academic and student data (Yohannes Kurniawan, Albertus Andika et al., 2019).

Managing student organizations shouldn't be a struggle. Suitable simplifies the management of student organizations by providing an intuitive platform for schools and students. It offers easy and stress-free ways to handle essential tasks, making the process more efficient and manageable (Visco, M. Jr. et al., 2024). The credibility and success of any organization, such as a college or university, are dependent on its capacity to provide secure, accurate, and timely data on its activities, stated by (Falebita Oluwatosin Samuel et al., 2022).

According to the study of (T.Liechty et al., 2020), As technology progresses, virtual communities are becoming more popular; however, research on virtual communities from a leisure standpoint is scarce. A study used netnography to investigate the meanings that members in a leisure-based virtual community assign to their experiences of community. The Hogwarts Running Club, a virtual club on Facebook, served as the study setting. But according to (Feng, S., Wong, Y. K., Wong, L. Y., & Hossain, L. et al., 2019), the excessive use of Facebook for entertainment has a negative effect on students' academic performance. However, (Baylee Patel et al., 2019) stated that educational website is a platform that allows students from all over the world to access courses, components, and services. Its major goal is to improve the learning experience by making education more accessible, convenient, and adaptable



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ages and backgrounds. Students who engaged in campus organizations relevant to their own cultural background developed networks of friends from many cultures, resulting in a stronger sense of belonging and loyalty to the institution (Krishna Bista, Chris Glass et al., 2019).

Synthesis

The review of relevant literature highlights the growing need for a comprehensive and efficient club administration system in educational institutions. Existing systems, as discussed by Cao and Chang (2023), often lack essential communication tools and role-specific features, leading to user dissatisfaction and delays in data retrieval. Vidya Pradosh et al. (2021) present an automated Cricket Club maintenance system, but its functionality is limited to specific tasks, failing to address broader administrative needs. Additionally, Liechty et al. (2020) identify a gap in research related to virtual communities, particularly in the context of leisure activities. Building on these findings, the proposed "eSAS - a Web-Based Student Organization Club Membership and Information System" aims to provide a more holistic solution that not only streamlines club management but also supports institutional goals of fostering student growth and engagement.

The introduction of eSAS addresses the critical demand for modernized and efficient management of student organizations within educational institutions. Studies by Balcita, Palaoag, and colleagues (2020) demonstrate



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that integrating technology into student services significantly enhances the quality of those services by automating key processes. The eSAS system replaces outdated manual methods with a centralized platform that improves student information and record management, while also facilitating seamless communication and coordination of activities (Srimathi R. et al., 2020). By providing a reliable, user-friendly platform, eSAS aligns with the broader institutional objective of increasing student participation and equipping students for future challenges. As noted by Chenjeri et al. (2024), the adoption of such technology not only enhances operational efficiency but also contributes to a supportive and dynamic campus environment. Consequently, eSAS represents a significant advancement in the management of student activities, aligning educational practices with contemporary technological developments.